

Database Design and Development

Week 01

Introduction to Database

Learning Outcomes

- At the end of the lesson you should be able to describe the fundamental concept of database design and database management

Prescribed Text

- Connolly, T. M. & Begg, C. E. (2005). Database Systems: A Practical Approach to Design, Implementation, and Management (4th Edition, Addison-Wesley Publishing. ISBN: 0321210255

Database Processing

- Traditional file processing Vs. Database processing.
- Database Processing :
 - All data is stored together
 - Special software is needed to process the data in the database
 - The database software is called Database management System (DBMS)
 - The DBMS provides capabilities for creating, accessing and updating data in a database

What is a Database System?

Database System = Database + DBMS

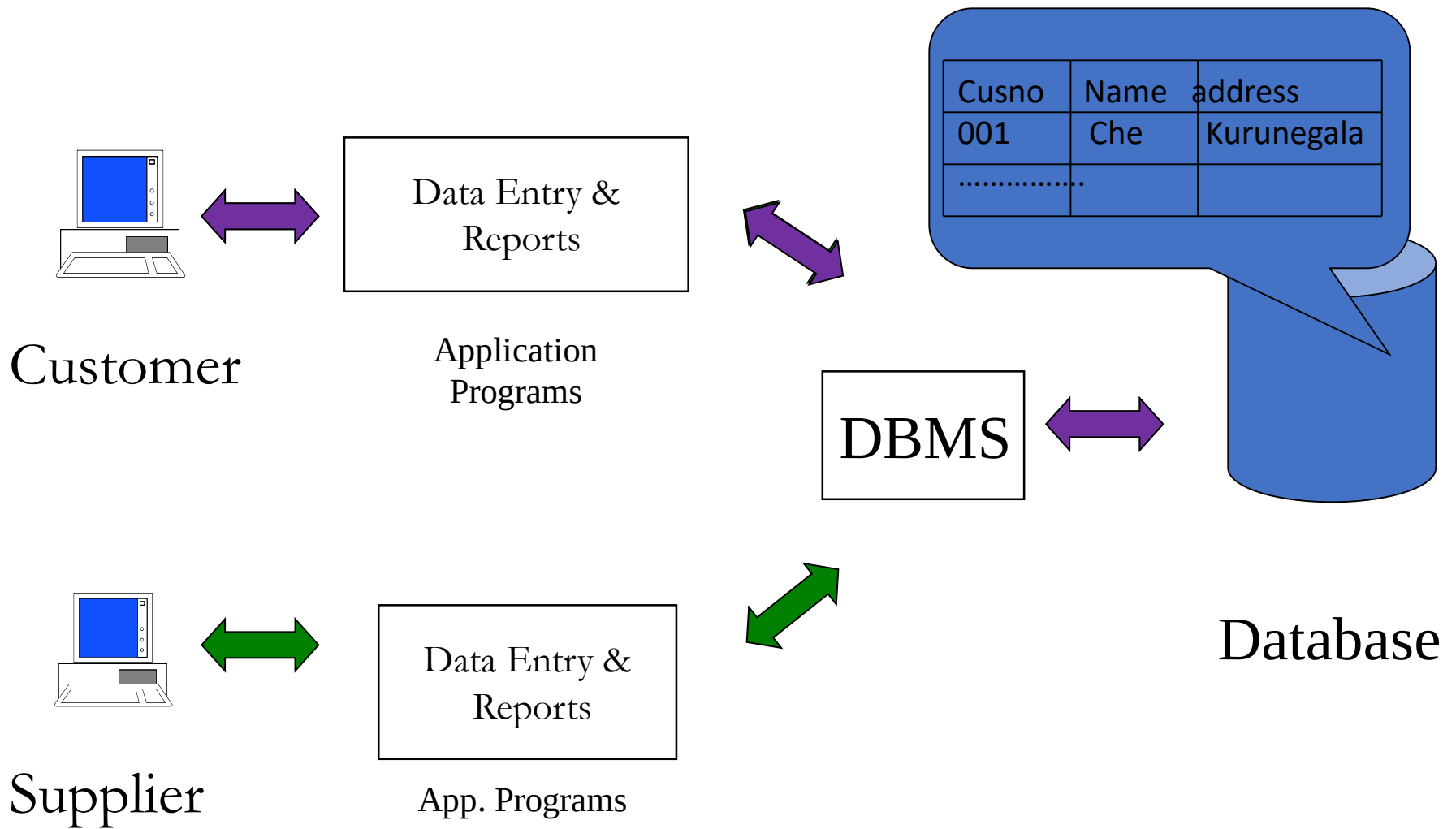
- A **Database** is
 - A large, integrated collection of data
 - Models (represents) a real-world enterprise.
- A **Database Management System (DBMS)** is
 - A software package designed to store and manages databases easily and efficiently.

Database

- Shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.

Database Management System

- A software system that enables users to define, create, maintain, and control access to the database



Database Applications

- **Banking**: all transactions
- **Airlines**: reservations, schedules
- **Universities**: registration, grades
- **Sales**: customers, products, purchases
- **Manufacturing**: production, inventory, orders, supply chain
- **Human resources**: employee records, salaries, tax deductions

Purpose of Database System

- Earlier database applications were built on top of traditional file systems
- Drawbacks of using file systems to store data:
 - Data redundancy and inconsistency
 - Difficulty in accessing data
 - Data isolation
 - Integrity problems
 - Poor enforcement of standards
 - Excessive program maintenance
 - Atomicity of updates
 - Concurrent access by multiple users
 - Security problems

Purpose of Database System

- ✓ Database systems offer solutions to all the above problems



Advantages of Database approach

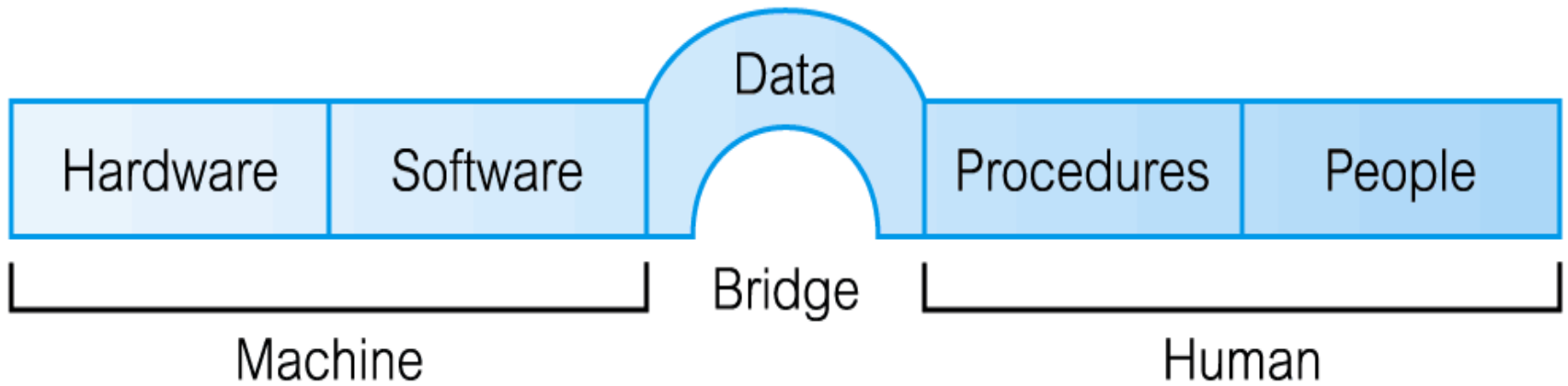
- Minimum data redundancy
- Consistency
- Integration of data
- Independence of data
- Data Sharing
- Enforcement of standards
- Ease of application development
- Uniform security, privacy and integrity control
- Flexibility, data accessibility and responsiveness
- Reduce program maintenance



Disadvantages of Database approach

- Introduces overhead on simple operations
- **Cost** and licensing issues
 - Do you need special hardware?
 - Can your customer base afford it?
- Learning Curve / **Training Issues**
- Need for specialized skills to maintain
- **Software Maintenance**
 - “Today’s software are tomorrow’s legacy applications”
- **Risk of data migration issues**
- Simplicity of relational data modeling (for RDBMS)
 - Do you need a DBMS to store your address book?

Components of DBMS Environment



Components of DBMS Environment

- Hardware

Can range from a PC to a network of computers.

- Software

DBMS, operating system, network software (if necessary) and also the application programs.

- Data

Used by the organization and a description of this data called the schema.

Components of DBMS Environment

- **Procedures**

Instructions and rules that should be applied to the design and use of the database and DBMS.

- **People**

Data Administrator (DA), Database Administrator (DBA), Database Designers (Logical and Physical), Application Programmers, End Users (naive and sophisticated)

Main Characteristics of the Database Approach

- Self-describing nature of a database system
- Insulation between programs and data
- Data Abstraction
- Support of multiple views of the data
- Data Sharing and multiuser transaction processing

History of Database Systems

- First-generation

- Hierarchical and Network

- Second generation

- Relational

- Third generation

- Object-Relational
- Object-Oriented

References

- Connolly, T. M. & Begg, C. E. (2005). Database Systems: A Practical Approach to Design, Implementation, and Management
- Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems